

NLP — Question Bank Answers

M1 — Detailed Exam-Oriented Answers

I'll provide the answers in a format you can **directly study and write in university exams**: definition → explanation → example → applications/advantages → code where useful.

Q1. Define NLP. Explain objectives and importance in Data Analytics.

Answer

Natural Language Processing (NLP) is a **branch of Artificial Intelligence (AI)** that enables computers to **understand, process, analyze, and generate human language in the form of text or speech.**

Human language is **unstructured and complex**. NLP converts this language into a form that computers can process and use for **analysis and decision-making**.

Objectives of NLP

The major objectives of NLP are:

1. **Understanding human language**
NLP helps computers understand the meaning and structure of human language.
2. **Text processing**
It processes large amounts of textual data such as emails, documents, reviews, and social-media posts.
3. **Information extraction**
Important information such as names, locations, dates, organizations, and events can be extracted.
4. **Text classification**
Documents can be classified into categories such as spam/not spam or positive/negative.
5. **Language generation**
NLP systems can generate human-readable text.
6. **Machine translation**
NLP can translate text from one language to another.
7. **Sentiment analysis**
It determines whether a statement expresses positive, negative, or neutral sentiment.

Importance in Data Analytics

A huge amount of modern data is **unstructured text**. NLP makes this data useful for analytics.

For example, consider thousands of customer reviews:

"The phone camera is excellent but the battery is poor."